

Java Syntax Essentials

Java syntax is the set of rules that defines how code must be written. Following these rules ensures the **compiler** understands your instructions.

1. The Core Structure: Classes

In Java, every line of executable code must reside inside a **class**. Think of a class as a blueprint or a container for your program.

```
public class Main { // This is a class
    // Code goes here
}
```

2. Naming & Case Sensitivity

Java is strictly **case-sensitive**. `Main` and `main` are viewed as two completely different things by the computer.

CORRECT:

File: `Main.java`

Class: `public class Main`

INCORRECT:

File: `Main.java`

Class: `public class main`

3. The Entry Point: `main()` Method

The `main()` method is the most important part of specialized Java files. It acts as the "Start Button" for your application.

```
public static void main(String[] args) { ... }
```

- **main:** The name the JVM looks for to start execution.
- **Curly Braces {}:** Mark the beginning and end of the block.

4. Printing Output

To see results on the screen, we use `System.out.println()`. Note the precise capitalization of **S**ystem.

```
System.out.println("This prints a line of text");
```

5. Semicolons: The "Full Stop"

Just as a sentence ends with a period, every Java statement must end with a **semicolon (;)**.

```
int x = 5; (Valid)
```

```
int x = 5 (Error: Semicolon missing)
```

6. Curly Braces { }

Braces are used to group code into blocks. If you open a brace `{`, you **must** close it `}`.

Pro-Tip: Always indent the code inside braces. It makes your program much easier to read!

7. Comments (Explaining Your Code)

Comments are ignored by the computer. They are written for humans to understand what the code does.

```
// This is a Single-line comment

/* This is a
   Multi-line
   comment */
```

Syntax Checklist

- Did I match my **File Name** to my **Class Name**?
- Did I capitalize **System** and **String**?
- Does every statement end with a **semicolon (;)**?
- Are all **Curly Braces** paired correctly?

Mastering syntax is the first step to writing bug-free code. Keep practicing!

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